

BISMUTH

(Data in metric tons gross weight unless otherwise noted)

Domestic Production and Use: The United States ceased production of primary refined bismuth in 1997 and is highly import dependent for its supply. Bismuth is contained in some lead ores mined domestically, but the last domestic primary lead smelter closed at yearend 2013, and all lead concentrates now are exported for smelting. In 2017, the estimated value of apparent consumption of bismuth was approximately \$22 million.

About two-thirds of domestic bismuth consumption was for chemicals used in cosmetic, industrial, laboratory, and pharmaceutical applications. Bismuth use in pharmaceuticals included bismuth salicylate (the active ingredient in over-the-counter stomach remedies) and other compounds used to treat burns, intestinal disorders, and stomach ulcers. Bismuth also is used in the manufacture of ceramic glazes, crystalware, and pearlescent pigments. Bismuth has a wide variety of metallurgical applications, including use as a nontoxic replacement for lead in brass, free-machining steels, and solders, and as an additive to enhance metallurgical quality in the foundry industry. The Safe Drinking Water Act Amendment of 1996, which required that all new and repaired fixtures and pipes for potable water supply be lead free after August 1998, opened a wider market for bismuth as a metallurgical additive to lead-free pipe fittings, fixtures, and water meters. Bismuth is used as a triggering mechanism for fire sprinklers and in holding devices for grinding optical lenses, and bismuth-tellurium-oxide alloy film paste is used in the manufacture of semiconductor devices.

Salient Statistics—United States:	2013	2014	2015	2016	2017^e
Production:					
Refinery	—	—	—	—	—
Secondary (old scrap) ^e	80	80	80	80	80
Imports for consumption, metal, alloys, and scrap	1,710	2,270	1,950	2,190	2,400
Exports, metal, alloys, and scrap	857	567	519	421	400
Consumption:					
Reported	774	727	662	733	770
Apparent ¹	1,010	1,500	1,380	1,670	2,080
Price, average, domestic dealer, dollars per pound	8.71	11.14	6.43	4.53	4.8
Stocks, yearend, consumer	50	329	464	643	640
Net import reliance ² as a percentage of apparent consumption	92	95	94	95	96

Recycling: Bismuth-containing new and old alloy scrap was recycled and thought to compose less than 5% of U.S. bismuth apparent consumption, or about 80 tons.

Import Sources (2013–16): China, 77%; Belgium, 12%; Peru, 3%; and other, 8%.

Tariff:	Item	Number	Normal Trade Relations 12–31–17
	Bismuth and articles thereof, including waste and scrap	8106.00.0000	Free.

Depletion Allowance: 22% (Domestic), 14% (Foreign).

Government Stockpile: None.

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Events, Trends, and Issues: The U.S. domestic dealer price of bismuth, which had trended upward in 2014, started 2015 at \$10.90 per pound, decreased steadily throughout the year, and ended the year with a December average of \$4.56 per pound. The price then remained relatively stable throughout 2016 and 2017, ranging between \$4.00 per pound and \$5.17 per pound.

In China, new environmental regulations were reported to have reduced the number of operating bismuth producers, from between 70 and 80 in 2016 to between 40 and 50 in 2017. Small- and medium-sized producers were forced to shut down for upgrades to meet stricter regulations, reducing domestic output by at least 25%.

World Refinery Production and Reserves: Mine production and reserves data were reevaluated, and available information was inadequate to make reliable estimates.

	Refinery production		Reserves ³
	2016	2017 ^e	
United States	—	—	Quantitative estimates of reserves are not available.
Canada	^e 25	25	
China	^e 14,000	11,000	
Japan	428	430	
Kazakhstan	140	140	
Laos	2,000	2000	
Mexico	539	540	
Russia	4	4	
World total (rounded)	17,100	14,000	

World Resources: Bismuth, at an estimated 8 parts per billion by weight, ranks 69th in elemental abundance in the Earth's crust and is about twice as abundant as gold. World reserves of bismuth are usually estimated based on the bismuth content of lead resources because bismuth production is most often a byproduct of processing lead ores. In China and Vietnam, bismuth production is a byproduct or coproduct of tungsten and other metal ore processing. Bismuth minerals rarely occur in sufficient quantities to be mined as principal products; the Tasna Mine in Bolivia and a mine in China are the only mines where bismuth has been the primary product. The Tasna Mine in Bolivia has been inactive since 1996.

Substitutes: Bismuth compounds can be replaced in pharmaceutical applications by alumina, antibiotics, and magnesia. Titanium dioxide-coated mica flakes and fish-scale extracts are substitutes in pigment uses. Indium can replace bismuth in low-temperature solders. Resins can replace bismuth alloys for holding metal shapes during machining, and glycerine-filled glass bulbs can replace bismuth alloys in triggering devices for fire sprinklers. Free-machining alloys can contain lead, selenium, or tellurium as a replacement for bismuth.

Bismuth is an environmentally friendly substitute for lead in plumbing and many other applications, including fishing weights, hunting ammunition, lubricating greases, and soldering alloys.

^eEstimated. — Zero.

¹Defined as secondary production + imports – exports + adjustments for industry stock changes.

²Defined as imports – exports + adjustments for industry stock changes.

³See [Appendix C](#) for resource and reserve definitions and information concerning data sources.